

TCP/IP Protocol Suite

1. TCP/IP Protocol Suite

The TCP/IP model is the conceptual model and set of communications protocols used in the Internet and similar networks. It has 4 layers: Application, Transport, Internet, and Link.

2. TCP vs UDP

Both are Transport layer protocols.

- TCP (Transmission Control Protocol): Connection-oriented, reliable, orders packets, error-checking, slower. Uses a three-way handshake (SYN, SYN-ACK, ACK).
- UDP (User Datagram Protocol): Connectionless, unreliable, unordered, faster, suited for streaming or gaming.

3. Port Numbers

Ports are logical constructs that identify a specific process or a type of network service. Standard ports include HTTP (80), HTTPS (443), FTP (20/21), SSH (22).

4. Key Application Layer Protocols

- HTTP (Hypertext Transfer Protocol): Used for transmitting web pages over the internet.
- FTP (File Transfer Protocol): Used for the transfer of computer files between a client and server.
- SMTP (Simple Mail Transfer Protocol): The standard protocol for email transmission.
- DNS (Domain Name System): Translates human-readable domain names (like www.google.com) to machine-readable IP addresses.
- DHCP (Dynamic Host Configuration Protocol): Automatically assigns IP addresses and other network configuration parameters to devices.